A1 Document Intelligence

Level: Easy. Plan for about 6 hours.

The customer

Kirana Logistics Group operates warehouses and trucking across Thailand, Indonesia, Vietnam and Malaysia. Its investor relations team answers the same analyst questions about the annual report every week. They want an assistant that answers questions from the report, shows where each answer comes from, and never makes things up.

What you get

- **Kirana_Logistics_Annual_Report_2025.pdf** (13 pages). It contains prose, tables, charts, an infographic, a diagram and a scanned page. As in most real reports, some facts appear only inside charts or images.

What to build (must have)

1. An **ingestion pipeline** that processes the whole PDF. If you choose not to handle some content type, say so and explain why.
2. A way to **ask questions**: CLI, API or simple web UI.
3. Answers that **cite their source**: page number at minimum, ideally the table, figure or passage.
4. **Honest refusals.** If the document does not support an answer, say so instead of guessing.
5. An **evaluation set** of 15 to 20 questions you write yourself, each with an expected answer and the page that supports it. Cover different content types, plus at least three questions the document cannot answer.
6. **Evaluation results** that measure retrieval (did the right evidence come back?) separately from answer quality (was the answer right?). Show the failures and what you think causes them.
7. The pipeline must work on a **new PDF without code changes**. In the demo round we will give you a second document from the same company.

Nice to have (pick any, or none)

- Hybrid search or reranking, with evidence that it helps
- Questions across several documents
- Cost and latency per question
- Conversation follow-ups
- A clean UI that shows the cited page

What we look for

Area	Strong submissions
Ingestion	Choose the right method per content type: text extraction, table parsing, OCR or a vision model. Explain the choice.
Chunking and retrieval	Can justify chunk size, overlap, metadata and top-k for this document.
Grounding	Verifiable citations; clear refusals when evidence is missing; conflicting statements surfaced, not hidden.

Area	Strong submissions
Evaluation	A repeatable test set, retrieval and answer quality measured separately, honest failure analysis.
Communication	README and customer summary a busy reader understands in five minutes.

Tip: start with a baseline that works end to end, measure it, then improve the weakest part. Show us that loop.

Take-home assignment: start here

Thank you for your time in the first conversation. The next step is a take-home assignment. There are three to choose from, and you do **one**. Each is modelled on a real kind of project our engineers deliver for customers in Southeast Asia. All companies, people and data in them are fictional.

The three options

	Assignment	Level	What it emphasises
A1	Document Intelligence: a question-answering assistant over an annual report	Easy	Sound RAG fundamentals, evaluation, honest answers
A2	Transaction Reconciliation: supplier invoices against an ERP export	Medium	Data integrity, deterministic logic, auditability
A3	Product Discovery: a sales assistant over a product catalog website	Hard	End-to-end design: data, search, agents, pricing

Pick the one where you can do your best work. A harder assignment earns no extra points; an excellent A1 beats an average A3. Harder assignments only give more room to show advanced skills.

Time

- Plan for about **6 hours** of focused work. Please do not spend much more; good judgment about scope is part of what we look at.
- You have **5 calendar days** from the day you receive this. The exact deadline (18:00 your local time) is in our email.
- If you need more time for any reason, including a holiday, tell us. Asking is fine.

What to submit

1. A Git repository link (GitHub or GitLab; private is fine if you give us access). Never commit API keys.
2. A **README** with: how to run it in 5 commands or fewer, an architecture overview, key decisions and trade-offs, known limitations, and what you would do with another week.
3. Your **evaluation results** as described in your brief, including cases where your system fails.
4. A **one-page customer summary** (PDF or Markdown) written for the customer's business owner, not an engineer: what you built, how well it works, what it cannot do yet, and what you recommend next.
5. One short README paragraph on **how you used AI tools** while building.

Submit by replying to the email we sent you, with your repository link and your one-page summary.

Rules

- Any language, framework, model provider or database. Free tiers and local models are fine.
- AI coding assistants are allowed. You must be able to explain and change any part of your code in the interview.
- If model API costs are a barrier, tell us and we will give you a capped API key.
- Use the data only for this assignment.
- Questions are welcome: reply to our email and we answer within one business day. Asking good clarifying questions is part of the job.

What happens after you submit

Round	Length	What happens
Demo and deep dive	60 min	You demo your work. We run it on inputs you have not seen. We discuss design, testing, failures and production readiness.
Customer simulation	40 min	A role-played discovery call, a short solution pitch and a conversation with a sceptical stakeholder.
Live engineering	45 min	A small debugging and diagnosis exercise. Bring your usual tools.

Keep your solution runnable for the demo round: we will ask you to process a new input file live, without changing code first.

How we evaluate

Every candidate is scored on the same seven areas, whichever assignment they pick: technical correctness, problem solving and judgment, engineering quality, AI and data evaluation, communication, business and pre-sales thinking, and ownership. We care more about evidence that your system works, and honesty about where it does not, than about polish.